



PlainBatteryGym: Building Reinforcement Learning Environments for Smart Battery-Charging Strategies

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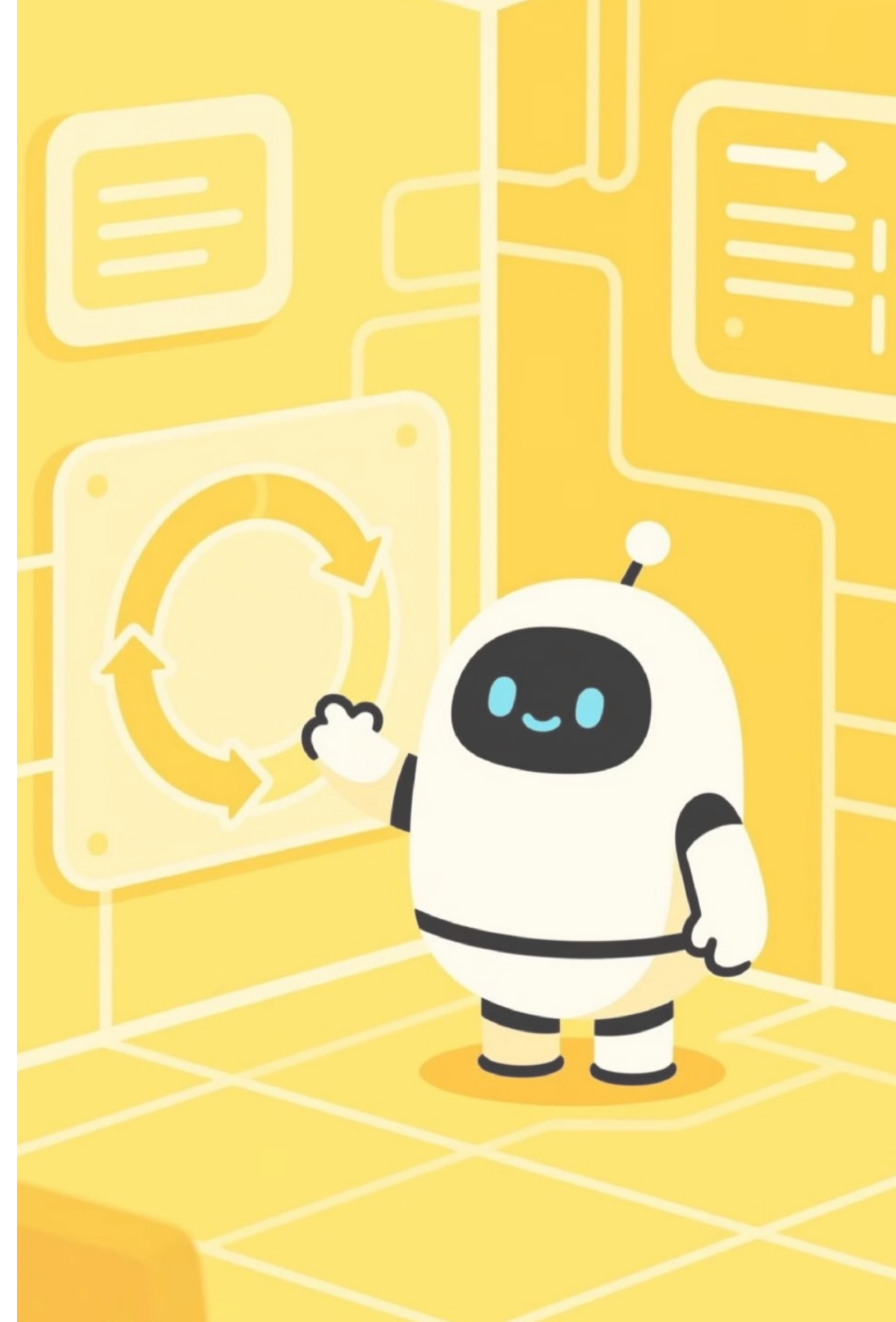
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The Problem:

RL Needs a Realistic Environment

- Battery control depends on household consumption, PV generation, electricity prices, and state of charge.
- RL agents need clearly defined observations, actions, rewards, and battery constraints.
- Manually building environments is slow, inconsistent, and hard to compare across experiments.

No realistic environment = no reliable agent training.





PlainBatteryGym Solution



Generate Gymnasium-compatible battery-control environments from clear plain specifications.



Plain specs. Smart batteries.
Reproducible RL.


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0.17 | 0.00 | 0.2911 | -0.29  
0.18 | 0.00 | 0.0364 | -0.04  
0.18 | 0.00 | -0.0629 | 0.06  
0.16 | 0.00 | 0.2906 | -0.29  
0.19 | 0.00 | 0.0340 | -0.03  
0.18 | 0.00 | -0.0626 | 0.06  
0.18 | 0.00 | 0.2910 | -0.29  
0.16 | 0.00 | 0.0388 | -0.04  
0.17 | 0.00 | -0.0644 | 0.06  
0.17 | 0.00 | 0.2889 | -0.29  
0.19 | 0.00 | 0.0360 | -0.04  
0.18 | 0.00 | -0.0627 | 0.06  
0.18 | 0.00 | 0.2904 | -0.29  
0.17 | 0.00 | 0.0394 | -0.04  
0.17 | 0.00 | -0.0636 | 0.06  
0.18 | 0.00 | 0.2880 | -0.29  
0.17 | 0.00 | 0.0385 | -0.04  
0.20 | 0.00 | -0.0640 | 0.06  
0.20 | 0.00 | 0.2951 | -0.30  
0.22 | 0.00 | 0.0428 | -0.04  
0.22 | 0.00 | -0.0607 | 0.06  
0.27 | 0.00 | 0.2986 | -0.30  
0.29 | 0.00 | 0.0560 | -0.06  
0.32 | 0.00 | -0.0564 | 0.06  
0.31 | 0.33 | 0.3247 | -0.32  
0.37 | 0.65 | 0.0012 | -0.00  
0.37 | 0.98 | -0.0916 | 0.09  
0.41 | 1.29 | 0.1340 | -0.13  
0.49 | 1.61 | -0.0518 | 0.05  
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```

```
ace/codeplain-energy/project$ python dist/examples/demo_battery_management.py --showcase --pv-kw 20 --hours 48
```

Thank You!

